

Robust nonlinear projection-based reduced-order models: comparative assessment of closure and manifold strategies

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Recent advances in nonlinear model reduction indicate that overcoming linear Kolmogorov limitations requires principled combinations of projection-based approximation and data-driven modeling [3]. In intrusive PROM settings, PROM-ANN introduced latent-space closure reconstruction of truncated modal coordinates [1], while PROM-RBF and PROM-GPR generalized this mechanism through alternative regression operators for the same closure channel [2]. In parallel, projection-compatible nonlinear latent-manifold formulations based on POD-autoencoders (POD-AE), in the spirit of POD-DL-ROM [4], provide an additional pathway to nonlinear approximation while preserving Galerkin/LSPG online dynamics.

The objective of this talk is to establish a coherent comparative framework for robust nonlinear PROM construction across these two families: closure-based reduced representations and manifold-based reduced representations. The analysis addresses central practical questions, including the selection of closure variables, sensitivity to high-mode reconstruction errors, interaction with intrusive residual minimization, and compatibility with hyper-reduction. To this end, we investigate metric-aware basis and closure variants together with computationally scalable intrusive formulations using ECSW hyper-reduction [5].

Numerical investigations are performed on a parametric two-component Burgers benchmark under both in-sample and held-out-parameter evaluations. The comparison is organized around three performance axes—approximation accuracy, computational cost, and achieved speed-up—with the goal of extracting practical guidance on how each strategy can be strengthened and in which operational settings it is most appropriate.

References

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